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“KHARKIV POLYTECHNIC INSTITUTE”
DEPARTMENT OF SOFTWARE ENGINEERING AND MANAGEMENT
INFORMATION TECHNOLOGIES”

Report of laboratory work № 2

Discipline «Software Engineering»

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Theme: Coding of Domain

Work objective: Learning basic principles of C++

Task:

$$13 \quad \left| y = \begin{cases} 13 - \prod_{i=4}^{n-3}(i+x), x < 0 \\ \prod_{j=1}^{n}\sum_{i=1}^{n+3} i^2(j + \frac{2-x}{i+4}), x \geq 0 \end{cases} \right|$$

Progress of work:

1. Writing a program based on the task from previous lab

1) Firstly, I created a code based on my previous lab task and Activity Diagram. The program is designed in C++ using an interactive console-based approach to perform mathematical formula calculations via wireframes. It utilizes specialized menu functions to separate user navigation, data input/output operations, and the core computational logic, ensuring code readability and ease of maintenance.

The program includes standard C++ libraries for console input/output (iostream), mathematical operations (cmath), file handling (fstream), text modifiers (termcolor.hpp), and system utilities. Two global variables, previous_results (string) and has_data (boolean), are defined to track the state of calculations and store data across different submenus. The Menu () function defines the welcome screen. It clears the terminal view, renders a framed title block with highlighted text, and prompts the user to press '1' to progress to the Main Menu. It also includes basic input validation. This part of the code is shown in Figure 1.

```cpp
#define TERMCOLOR_USE_ANSI_ESCAPE_SEQUENCES
#include <iostream>
#include <cmath>
#include <windows.h>
#include "termcolor.hpp"
#include <conio.h>
#include <string>
#include <fstream>
using namespace std;
using namespace termcolor;

string previous_results = "";
bool has_data = false;



int Menu()
{
    char choice = 0;
    system("cls");
    cout << "=================================================" << endl;
    cout << "                                                 " << endl;
    cout << green << "            Formula calculation program          " << reset << endl;
    cout << "                                                 " << endl;
    cout << "=================================================" << endl;
    cout << "                                                 " << endl;
    cout << "                                                 " << endl;
    cout << "         Welcome to the formula calculating      " << endl;
    cout << "                    program.                     " << endl;
    cout << "                                                 " << endl;
    cout << "         Press 1 to go to the Main menu          " << endl;
    cout << "                                                 " << endl;
    cout << "      *Read `Help` before starting the program*  " << endl;
    cout << "                                                 " << endl;
    cout << "=================================================" << endl;
        cin >> choice;
    if (choice != '1')
    {
        cout << "Invalid input. Please try again." << endl;
        system("pause");
        return Menu();
    }
    return 1;
}
```

Figure 1. Implemented libraries and greeting menu function

2) The MainMenu() function displays the program's capabilities (performing calculations, processing results, providing guidance on how the program works and its underlying logic, and exiting). The user is presented with a choice from 1 to 4, corresponding to the category they wish to interact with.

```cpp

int MainMenu()
{
        char choice = 0;
        system("cls");
        cout << "=================================================" << endl;
        cout << "                                                 " << endl;
        cout << green << "                      Main menu          " << reset << endl;
        cout << "                                                 " << endl;
        cout << "=================================================" << endl;
        cout << "                                                 " << endl;
        cout << "                                                 " << endl;
        cout << "             1. Run calculations                 " << endl;
        cout << "                                                 " << endl;
        cout << "             2. Management of data               " << endl;
        cout << "                                                 " << endl;
        cout << "             3. Help                             " << endl;
        cout << "                                                 " << endl;
        cout << "             4. Exit the program                 " << endl;
        cout << "                                                 " << endl;
        cout << "=================================================" << endl;
        cout << "                                                 " << endl;
        cout << "             select an option: ";
        cin >> choice;

        if (choice < '1' || choice > '4')
        {
            cout << "Invalid input. Please try again." << endl;
            system("pause");
            return MainMenu();
        }

        return choice - '0';
}
```

Figure 2. Main menu

3) CalculationsMenu() function is the main part of the program. It contains a function calculator. The user is given the option to input variable data either via a file (runFileCalculations()) or through the keyboard(runCalculations()).

```
79
80 int CalculationsMenu()
81 {
82     char choice = 0;
83     system("cls");
84     cout << "=====================================================" << endl;
85     cout << "                                                     " << endl;
86     cout << green << "                Run calculations           " << reset << endl;
87     cout << "                                                     " << endl;
88     cout << "=====================================================" << endl;
89     cout << "                                                     " << endl;
90     cout << "                                                     " << endl;
91     cout << "             1. Input from a file                     " << endl;
92     cout << "                                                     " << endl;
93     cout << "             2. Input from a keyboard                " << endl;
94     cout << "                                                     " << endl;
95     cout << "                                                     " << endl;
96     cout << "       ---------------------------------             " << endl;
97     cout << "                                                     " << endl;
98     cout << "     press 3 to return to the Main menu              " << endl;
99     cout << "                                                     " << endl;
100    cout << "=====================================================" << endl;
101    cout << "                                                     " << endl;
102    cout << "             select an option : ";
103    cin >> choice;
104
105    if (choice == '3')
106    {
107        return '3';
108    }
109
110    if (choice < '1' || choice > '2')
111    {
112        cout << "Invalid choice. Please try again." << endl;
113        system("pause");
114        return CalculationsMenu();
115    }
116 return choice;
117 }
```

Figure 3. Run calculations

```
182
183 char runFileCalculations()
184 {
185     has_data = true;
186     system("cls");
187     cout << "=====================================================" << endl;
188     cout << endl;
189     cout << "       Select a file from your device" << endl;
190     cout << "      (supported formats .txt, .csv, .xlsx)" << endl;
191     cout << endl;
192     cout << "-----------------------------------------------------" << endl;
193     cout << endl;
194     cout << "    Press q to return to the previous menu    " << endl;
195     cout << endl;
196     cout << "-----------------------------------------------------" << endl<< endl;
197     cout << "    Press 3 to return to the Main menu" << endl<< endl;
198     cout << "=====================================================" << endl<< endl;
199     cout << "    Enter file name (e.g. data.txt) or other option: ";
200
201     string filename;
202     cin >> filename;
203
204     if (filename == "3")
205     {
206         return '3';
207     }
208     if (filename == "q" || filename == "Q")
209     {
210         return 'q';
211     }
212
213     ifstream file(filename);
214     if (!file.is_open())
215     {
216                 cout << endl;
217         cout << red << "Error: Could not open file! File might not exist." << reset << endl;
218         system("pause");
219         return runFileCalculations();
220     }
221
```

```
    unsigned int n;
    float a, b, x, h;

    if (!(file >> n))
    {
        cout << endl << red << "Error in file: Invalid 'n' parameter! (Must be an integer >= 7, no letters/symbols)" << reset << endl;
        file.close(); system("pause");
        return 0;
    }
    if (n < 7)
    {
        cout << endl << red << "Error in file: 'n' parameter must be >= 7 (Got: " << n << ")" << reset << endl;
        file.close(); system("pause");
        return 0;
    }
    if (!(file >> h))
    {
        cout << endl << red << "Error in file: Invalid 'h' step! (Must be a number > 0, no letters/symbols)" << reset << endl;
        file.close(); system("pause");
        return 0;
    }
    if (h <= 0)
    {
        cout << endl << red << "Error in file: 'h' step must be > 0 (Got: " << h << ")" << reset << endl;
        file.close(); system("pause");
        return 0;
    }
    if (!(file >> a))
    {
        cout << endl << red << "Error in file: Invalid 'a' lower bound! (Must be a number, no letters/symbols)" << reset << endl;
        file.close(); system("pause");
        return 0;
    }
    if (!(file >> b))
    {
        cout << endl << red << "Error in file: Invalid 'b' upper bound! (Must be a number, no letters/symbols)" << reset << endl;
        file.close(); system("pause");
        return 0;
    }
    if (a > b)
    {
        cout << endl << red << "Error in file: Lower bound 'a' cannot be greater than upper bound 'b' (a <= b)" << reset << endl;
        file.close(); system("pause");
        return 0;
    }
```

```
268     file.close();
269
270     system("cls");
271     cout << "=====================================================" << endl << endl;
272     cout << cyan << "File \"" << filename << "\" added successfully!" << reset<< endl<< endl;
273     cout << "Processing the data..." << endl << endl;
274     cout << "-----------------------------------------------------" << endl;
275     cout << endl << endl << endl;
276     cout << "Results for data from " << filename << ":" << endl << endl;
277
278     for (x = a; x <= b; x += h)
279     {
280         float y = 0.0;
281         float product = 1.0;
282         if (x < 0) {
283             for (int i = 4; i <= (int)n - 3; i++) { product *= (i + x); }
284             y = 13 - product;
285         }
286         else {
287             for (int j = 1; j <= n; j++) {
288                 float sum = 0.0;
289                 for (int i = 1; i <= (int)n + 3; i++) {
290                     sum += pow(i, 2) * (j + (2 - x) / (i + 4));
291                 }
292                 product *= sum;
293             }
294             y = product;
295         }
296         cout << "x = " << x << "  |  y = " << y << endl;
297         previous_results += "x = " + to_string(x) + "  |  y = " + to_string(y) + "\n";
298
299     }
300    return askToSaveFile();
301 }
```

Figure 3.1 Input from a file

For input validation, I used do-while statements to make sure the user enters the correct values within the specified range. Then, I divided my function into two parts for different intervals using an if statement. Inside these sections, for loops are used to calculate the product of sums and the product itself based on the increments of i and j. Finally, the values of variables x and y are displayed for each step within the interval from a to b.

```
481  char runCalculations()
482  {
483      char choice = 0;
484      has_data = true;
485      system("cls");
486      unsigned int n;
487      float a, b, x, h;
488      cout << endl;
489      cout << cyan << "                                MATHEMATICAL FORMULAE                          " << reset << endl;
490      cout << endl;
491      cout << endl;
492      cout << "   Depending on 'x' value, the program uses two variants:   " << endl;
493      cout << endl;
494      cout << "   1) If x < 0:" << endl;
495      cout << "      Math form: y = 13 - P_{i=4}^{n-3} (i + x)             " << endl;
496      cout << endl;
497      cout << "   2) If x >= 0:" << endl;
498      cout << "      y = P_{j=1}^{n} [ S_{i=1}^{n+3} (i^2 * (j + (2 - x) / (i + 4))) ]" << endl;
499      cout << endl;
500      cout <<         "-------------------------------------------------------------------------" << endl;
501      cout << endl;
502      cout << "                   Press q to return to the previous menu    " << endl;
503      cout << endl;
504      cout << endl << "-------------------------------------------------------------------------" << endl;
505      cout << endl;
506      cout << "                   Press 3 to return to the Main menu        " << endl;
507      cout << endl;
508      cout << endl << "-------------------------------------------------------------------------" << endl;
509      cout << endl;
510      cout << "                   Press any key to start calculations    " << endl;
511      cout << endl;
512      cout << endl << "-------------------------------------------------------------------------" << endl;
513      cout << endl;
514      cout << "select an option:";
515      cin >> choice;
516      cout << endl;
```

```
518      if (choice == '3')
519      {
520          return '3';
521      }
522          else if (choice == 'q'||choice == 'Q')
523          {
524                  return 'q';
525          }
526      else
527      {
528          system("pause > nul");
529      }
530      cout << "=========================================================================" << endl
531
532      do
533      {
534          cout << endl;
535          cout << "Input n - parameter (n>=7): ";
536          cin >> n;
537          if (cin.fail())
538          {
539              cin.clear();
540              cin.ignore(10000, '\n');
541              cout << red << "Error: Please enter a valid number!" << reset << endl;
542              n = 0;
543          }
544      } while (n < 7);
545      do
546      {
547          cout << "Input h - step (h>0): ";
548          cin >> h;
549          if (cin.fail())
550          {
551              cin.clear();
552              cin.ignore(10000, '\n');
553              cout << red << "Error: Please enter a valid number!" << reset << endl;
554              h = 0.0f;
555          }
556      } while (h <= 0);
```

```
557     do
558     {
559         cout << "Input a - lower bound of the interval: ";
560         cin >> a;
561         if (cin.fail())
562         {
563             cin.clear();
564             cin.ignore(10000, '\n');
565             cout << red << "Error: Invalid input for 'a'!" << reset << endl;
566             a = 1; b = 0;
567             continue;
568         }
569         cout << "Input b - upper bound of the interval(a <= b): ";
570         cin >> b;
571         if (cin.fail())
572         {
573             cin.clear();
574             cin.ignore(10000, '\n');
575             cout << red << "Error: Invalid input for 'b'!" << reset << endl;
576             a = 1; b = 0;
577             continue;
578         }
579         cout << endl;
580     } while (a > b);
581     cout << "========================================================================" << endl;
582     for (x = a; x <= b; x += h)
583     {
584         float y = 0.0;
585         float product = 1.0;
586         if (x < 0)
587         {
588             for (int i = 4; i <= n - 3; i++)
589             {
590                 product *= (i + x);
591             }
592             y = 13 - product;
593         }
594         else
595         {
596             for (int j = 1; j <= n; j++)
597             {
598                 float sum = 0.0;
599                 for (int i = 1; i <= n + 3; i++)
600                 {
601                     sum += pow(i, 2) * (j + (2 - x) / (i + 4));
602                 }
603                 product *= sum;
604             }
605             y = product;
606         }
607         cout << endl << "x= " << x << "\ny= " << y << endl;
608         previous_results += "x = " + to_string(x) + "  |  y = " + to_string(y) + "\n";
609     }
610     return askToSaveFile();
611 }
```

Figure 3.2 Input from a keyboard

After completing the calculations, the user is given the opportunity to save the results in a .txt file. The function askToSaveFile() performs an initial check to ensure calculation records exist before proceeding.

```
120  char askToSaveFile()
121  {
122      if (previous_results == "")
123      {
124          return 0;
125      }
126      cout << endl;
127      cout << endl;
128      cout << "       Do you want to save the result in a file?" << endl << endl;
129      cout << endl;
130      cout << "                       1. Yes" << endl;
131      cout << "                       2. No" << endl;
132      cout << endl;
133      cout << "-------------------------------------------------" << endl;
134      cout << endl;
135      cout << "       Press 3 to return to the Main menu" << endl;
136      cout << endl;
137      cout << "=================================================" << endl;
138      cout << endl;
139      cout << "              select an option: ";
140
141      char choice;
142      cin >> choice;
143
144      if (choice == '1')
145      {
146          system("cls");
147          string outFileName = "results.txt";
148          ofstream outFile(outFileName);
149          if (outFile.is_open())
150          {
151              outFile << previous_results;
152              outFile.close();
153
154              cout << "=======================================" << endl << endl;
155              cout << cyan << "File \"" << outFileName << "\" saved successfully!"<< reset << endl<<endl;
156              cout << "---------------------------------------" << endl;
157              cout << endl;
158              cout << "Press any key to return to the previous menu..." << endl;
159              cout << endl;
160              cout << "=======================================" << endl;
161              system("pause > nul");
```

```
162          }
163          else
164          {
165              cout << endl;
166              cout << "Error: Could not save the file!" << endl;
167              cout << endl;
168          }
169      }
170      else if (choice == '2')
171      {
172          return '2';
173      }
174      else if (choice == '3')
175      {
176          system("cls");
177
178          return '3';
179      }
180      return 0;
181  }
```

Figure 4. Saving results in a file

4) Data Menu function is responsible for handling previously calculated data. It enables users to display stored results and delete existing records. If no calculations have been performed, the program informs the user that no results are available. In other case it gives an option to save them to a file (Fig. 4), Before clearing data, the user is asked to confirm the action. Navigation options are provided to return to previous menus or the main menu, ensuring convenient interaction with the program.

```
304 int DataMenu()
305 {
306     char choice = 0;
307     system("cls");
308     cout << "================================================" << endl;
309     cout << "                                                " << endl;
310     cout << green << "             Management of data              " << reset << endl;
311     cout << "                                                " << endl;
312     cout << "================================================" << endl;
313     cout << "                                                " << endl;
314     cout << "                                                " << endl;
315     cout << "         1. Show the previous result            " << endl;
316     cout << "                                                " << endl;
317     cout << "         2. Clear previous data                 " << endl;
318     cout << "                                                " << endl;
319     cout << "                                                " << endl;
320     cout << "      --------------------------------          " << endl;
321     cout << "                                                " << endl;
322     cout << "     press 3 to return to the Main menu         " << endl;
323     cout << "                                                " << endl;
324     cout << "================================================" << endl;
325     cout << "                                                " << endl;
326     cout << "            select an option : ";
327
328     cin >> choice;
329
330     if (choice == '1' || choice == '2' || choice == '3')
331     {
332         return choice - '0';
333     }
334     else
335     {
336         cout << "Invalid choice. Please try again." << endl;
337         system("pause");
338         return DataMenu();
339     }
340 }
```

Figure 5. Management of data

```
670                 {
671                     system("cls");
672                     cout << "================================================" << endl;
673                     cout << endl;
674                     if (!has_data || previous_results == "")
675                     {
676                         cout << "No previous data found. Run calculations first!" << endl;
677                         cout << endl;
678                         cout << "------------------------------------------------" << endl;
679                         cout << endl;
680                         cout << "Press any key to return...";
681                         system("pause > nul");
682                     }
683                     else
684                     {
685                         cout << previous_results;
686                         char result = askToSaveFile();
687                         if (result == '3')
688                         {
689                             break;
690                         }
691                     }
```

Figure 5.1. Showing the results

```
344  int clearData()
345  {
346      system("cls");
347      cout << "====================================" << endl;
348      cout << endl;
349      cout << "Are you sure you want to clear the data?" << endl;
350      cout << endl;
351      cout << red << "!Warning: This action will delete your"<< endl << "    results permanently!" << reset << endl;
352      cout << endl;
353      cout << "              1. Yes" << endl;
354      cout << "              2. No" << endl;
355      cout << endl;
356      cout << "====================================" << endl;
357      cout << endl;
358      cout << "          select an option: ";
359
360      char choice;
361      cin >> choice;
362      if (choice == '1')
363      {
364          system("cls");
365          cout << "------------------------------------" << endl;
366          cout << endl;
367          cout << "      Deleting the data..." << endl;
368          cout << endl;
```

```
370              previous_results = "";
371              has_data = false;
372
373              cout << cyan << "     Data cleared successfully!" << endl << reset;
374          }
375          else if (choice == '2')
376          {
377              return 0;
378          }
379          else
380          {
381              cout << "Invalid input. Please try again." << endl;
382              system("pause");
383              return clearData();
384          }
385          cout << "------------------------------------" << endl;
386          cout << endl;
387          cout << "       Press any key to return..." << endl;
388          cout << endl;
389          cout << "====================================" << endl;
390          system("pause > nul");
391          return 0;
392      }
```

Figure 5.2. Clearing of data

5) The last two functions operate in a rather simplified manner compared to the others. The Help () function provides the user with the information needed to carry out further calculations or to work with the program itself.

When the option to exit the program in function Exit () is selected, the user is asked to confirm this action, after which they can safely close the program.

```cpp
int HelpMenu()
{
    char choice = 0;
    system("cls");
    cout << "=================================================================" << endl;
    cout << "                                                                 " << endl;
    cout << green << "                              Help                               " << reset << endl;
    cout << "                                                                 " << endl;
    cout << "=================================================================" << endl;
    cout << "PURPOSE OF THE PROGRAM:                                          " << endl;
    cout << "   This program is designed to calculate the values of a         " << endl;
    cout << "   function according to formulae for a given range of data.     " << endl;
    cout << endl;
    cout << "1. DATA INPUT OPTIONS:                                           " << endl;
    cout << "   - keyboard                                                    " << endl;
    cout << "   - file                                                        " << endl;
    cout << endl;
    cout << "2. DATA MANAGEMENT OPTIONS:                                      " << endl;
    cout << "   - View: See the calculated results directly on the screen     " << endl;
    cout << "   - Save: Store your results into an output file                " << endl;
    cout << "   - Delete: Clear previous calculation data                     " << endl;
    cout << endl;
    cout << "MATHEMATICAL FORMULAE:                                           " << endl;
    cout << "   Depending on 'x' value, the program uses two variants:        " << endl;
    cout << endl;
    cout << "   1) If x < 0:" << endl;
    cout << "      Math form: y = 13 - P_{i=4}^{n-3} (i + x)                  " << endl;
    cout << endl;
    cout << "   2) If x >= 0:" << endl;
    cout << "      y = P_{j=1}^{n} [ S_{i=1}^{n+3} (i^2 * (j + (2 - x) / (i + 4))) ]" << endl;
    cout << "=================================================================" << endl;
    cout << endl;
    cout << "         press 3 to return to the Main menu                      " << endl;

    cin >> choice;
    if (choice == '3')
    {
        return '3';
    }
 return 0;
}
```

Figure 6. Help

```cpp
int ExitMenu()
{
    char choice = 0;
    system("cls");
    cout << "=======================================" << endl;
    cout << endl;
    cout << "      Are you sure you want to exit?     " << endl;
    cout << endl;
    cout << "           1. Yes (1)" << endl;
    cout << "           2. No (2)" << endl;
        cout << endl;
    cout << "---------------------------------------" << endl;
    cout << endl;
    cout << "Press 3 to return to the Main menu      " << endl;
    cout << endl;
    cout << "=======================================" << endl;
    cout << endl;
    cout << "       select an option:";

    cin >> choice;

    if (choice == '1')
    {
        system("cls");
        cout << endl;
                cout << "-----------------------------------" << endl;
                cout << endl;
        cout << "      Exiting the program..." << endl << endl;
                cout << "-----------------------------------" << endl;
        return 1;
    }
    if (choice == '2')
    {
        return 0;
    }

    if (choice == '3')
    {
       return 0;
    }
    cout << "Invalid input. Please try again." << endl;
    system("pause");
    return ExitMenu();
}
```

Figure 7. Exit

7)The global runtime process is led by the main () function, which acts as the primary entry point of the software. At startup, it configures the Windows console environment to process virtual terminal attributes for text formatting. Subsequently, main () enters a persistent execution loop embedded with a multi-branch switch statement. It monitors system state outputs, coordinates transitions between submenus, processes data flow threads, and guarantees a stable program termination once the exit parameters are fulfilled.

```
613 int main()
614 {
615     HANDLE hOut = GetStdHandle(STD_OUTPUT_HANDLE);
616     DWORD dwMode = 0;
617     GetConsoleMode(hOut, &dwMode);
618     SetConsoleMode(hOut, dwMode | ENABLE_VIRTUAL_TERMINAL_PROCESSING);
619
620     Menu();
621     bool running = true;
622     while (running)
623     {
624         switch (MainMenu())
625         {
626         case 1:
627             while (true)
628             {
629                 char calc = CalculationsMenu();
630                 if (calc == '3')
631                 {
632                     break;
633                 }
634                 if (calc == '1')
635                 {
636                     char result = runFileCalculations();
637                     if (result == '3')
638                     {
639                         break;
640                     }
641                     if (result == 'q' || result == 'Q')
642                     {
643                         continue;
644                     }
645                 }
646                 else if (calc == '2')
647                 {
648                     char result = runCalculations();
649                     if (result == '3')
650                     {
651                         break;
652                     }
653                     if (result == 'q' || result == 'Q')
654                     {
655                         continue;
656                     }
657
658                 }
659             }
```

```
660             break;
661         case 2:
662             while (true)
663             {
664                 int dataman = DataMenu();
665                 if (dataman == 3)
666                 {
667                     break;
668                 }
669                 if (dataman == 1)
670                 {
671                     system("cls");
672                     cout << "================================================" << endl;
673                     cout << endl;
674                     if (!has_data || previous_results == "")
675                     {
676                         cout << "No previous data found. Run calculations first!" << endl;
677                         cout << endl;
678                         cout << "------------------------------------------------" << endl;
679                         cout << endl;
680                         cout << "Press any key to return...";
681                         system("pause > nul");
682                     }
683                     else
684                     {
685                         cout << previous_results;
686                         char result = askToSaveFile();
687                         if (result == '3')
688                         {
689                             break;
690                         }
691                     }
692                 }
693                 else if (dataman == 2)
694                 {
695                      clearData();
696                                 }
697             }
698             break;
699         case 3:
700             HelpMenu();
701             break;
```

```
702         case 4:
703             if (ExitMenu() == 1)
704             {
705                 running = false;
706             }
707             break;
708 
709         }
710 
711     }
712     return 0;
713 }
```

Figure 8. Main function

2. Enhancing interface and user experience of console application. Creating a wireframe to visualize the idea

While designing the wireframes, I focused on creating a simple and intuitive user experience. Each screen contains only the controls and information required for the current task, reducing unnecessary complexity.

The navigation is consistent across all screens, allowing the user to easily return to the Main menu by pressing the 3 key. Confirmation screens are used before important actions such as deleting data or exiting the program to prevent accidental operations. The Help screen provides clear instructions about the program's functionality, while the Results and Data management screens allow users to view, save, and manage calculated results efficiently. This design helps user complete tasks quickly with minimal effort and confusion.

The wireframes are shown in Figure 9. The flowchart is shown in Figure 10.



```
========================================

          Management of data

========================================

       1. Show the previous result

         2. Clear previous data


   ----------------------------------

    press 3 to return to the Main menu

========================================

          select an option: _
```

```
========================================

        [Display of the results]


Do you want to save the result in a file?

                1.Yes
                2. No

----------------------------------------

    press 3 to return to the Main menu

========================================

           select an option:
```

```
==================================================

                    Help

==================================================

PURPOSE OF THE PROGRAM:
This program is designed to calculate the values of a
function according to formulae for a given range of
data.

1. DATA INPUT OPTIONS:
   - keyboard
   - file

2. DATA MANAGEMENT OPTIONS:
   - View: See the calculated results directly on the
screen
   - Save: Store your results into an output file
   - Delete: Clear previous calculation data

MATHEMATICAL FORMULAE:

Depending on 'x' value, the program uses two variants:

1) If x < 0:
Math form: y = 13 - P_{i=4}^{n-3} (i + x)
2) If x >= 0:
y = P_{j=1}^{n} [ S_{i=1}^{n+3} (i^2 * (j + (2 - x) /
(i + 4))) ]

==================================================

press 3 to return to the Main menu
```

```
========================================

    Are you sure you want to exit?

              Yes(1)
              No(2)


          Exiting the program...


----------------------------------

    press 3 to return to the Main menu

========================================
```

```
                MATHEMATICAL FORMULAE:

Depending on 'x' value, the program uses two variants:

1) If x < 0:
y = 13 - [(4 + x) * (5 + x) * ... * ((n - 3) + x)]
2) If x >= 0:
y = P_{j=1}^{n} [ S_{i=1}^{n+3} (i^2 * (j + (2 - x) / (i + 4))) ]

--------------------------------------------------------------------------

                Press q to return to the previous menu

--------------------------------------------------------------------------

                Press 3 to return to the Main menu

--------------------------------------------------------------------------

                Press any key to start calculations

--------------------------------------------------------------------------

select an option:


==========================================================================

Input n (n>=7): _
Input h (h>0): _
Input a: _
Input b (a <= b):


==========================================================================
```

```
========================================

    Are you sure you want to clear
            the data?

  !Warning: This action will delete your
          results permanently!

              1. Yes
              2. No

========================================

    press 3 to return to the Main menu

========================================


----------------------------------
        Deleting the data...

      Data cleared successfully!
----------------------------------

     Press any key to return...

========================================
```

Figure 9. Wireframes

Based on the created wireframes, I developed a flowchart that represents the main functionality of the application and shows how users can navigate between different screens:

Use Case 1: Run calculations

This use case allows the user to calculate the function values. The user can choose to enter input data either from the keyboard or from a file. After the required data is provided, the program performs the calculations and redirects the user to the Display of results screen.

Use Case 2: Input from a keyboard

This use case allows the user to manually enter all required parameters. Can redirect the user back to the previous menus.

Use Case 3: Input from a file

This use case allows the user to load previously prepared input data from a file. Can redirect the user back to the previous menus.

Use Case 4: Display of results

This use case presents the calculated values to the user. It can be reached either after entering data from the keyboard, loading data from a file, or selecting the previous result.

Use Case 5: Save to file

This use case allows the user to save the displayed results to an external file.

Use Case 6: Help

This use case provides information about the purpose of the application and explains its main functionality. It helps users understand how the system works.

Use Case 7: Management of data

This use case groups all operations related to working with previously generated results. It contains two options: Show previous result and Clear data.

Use Case 8: Show previous result

This use case allows the user to view the most recently generated result. It leads to the Display of results screen.

Use Case 9: Clear data

This use case enables the user to permanently remove previously saved results from the program`s data. A confirmation screen is displayed before the operation is performed to prevent accidental data loss.

Use Case 10: Exit

This use case allows the user to terminate the program. Before closing the program, the user is asked to confirm the action. If confirmed, the application is closed, otherwise, the user returns to the Main menu.

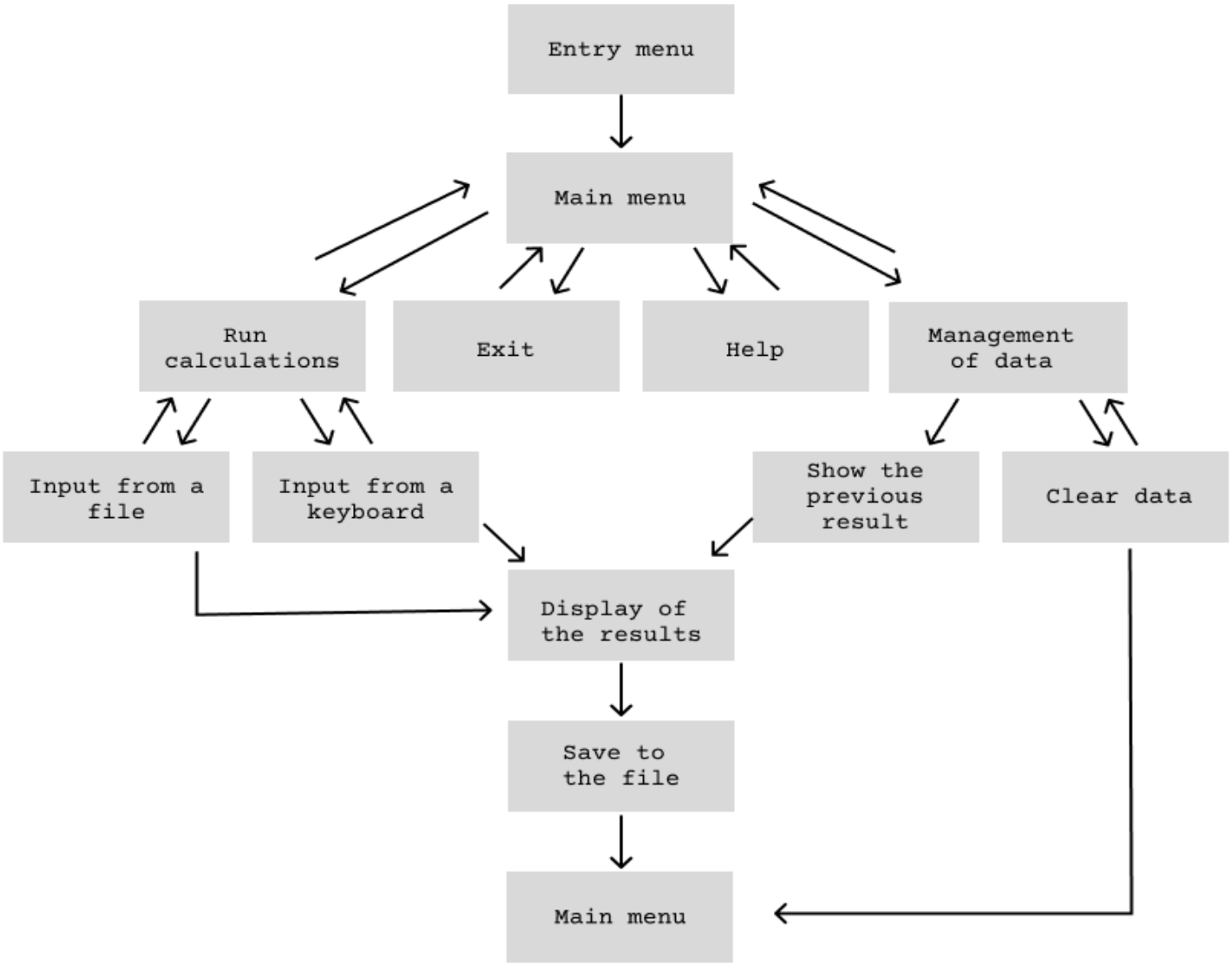


Figure 10. Flowchart diagram

Conclusions: During execution of this laboratory training, I gained knowledge and skills in developing program code and wireframes. I described the algorithms for calculating sums, products, and sums of products using loops and nested loops while using my knowledge about statements and basics of C++ programming in VS. I also created a graphical representation of my program through wireframes in Figma. The laboratory work helped me practice my programming skills and added a clearer explanation to the UX design.